S²-MAS

Scaling persona-driven LLM simulation

A society-scale multi-agent simulation framework with 500 million household persona agents

500M
synthetic household personas

2,925,263
largest finance set in paper comparison

12M
distinct model-driven household agents

14 GPUs
3 h 59 min end to end

12M households → 24M paired-condition choices → 288M monthly ledger states

8 H100 80GB + 6 A100 40GB

Data
Real-survey data
500M personas · 36 coded fields

→

sharp decision models
SFT + RLCD · decision heads · LoRA
Full state + complete alternatives

→

Genese engine
World state ↔ Brain policy intents
Batched arrays · state–event commit

→

Financial experiments
8 reusable task families
Consumption · cash · repayment

Consumer financial choices

Model	Accuracy (%)	Q-macro F1 (%)
sharp-421m-mlx	**74.55**	**38.37**
sharp-4b	74.14	36.73
sharp-9b	74.09	36.66
GLM-5.3-Flash	68.88	38.00
Jev 1.13	65.26	37.48

240,458 shared test records · 119 question versions

421m MLX FP16; 4b/9b BF16; API baselines

Six approaches, one sharp family

sharp-421m
421M · English · full-data SFT + RLCD

sharp-322m
322M · bilingual · PyTorch / MLX

sharp-4b
4B · head-only / LoRA + head

sharp-9b
9B · LoRA + decision head

6 post-training recipes · 10 trained variants

SFT · SFT→CE · SFT+RLCD · 4B head

4B LoRA+head · 9B LoRA+head

sharp-322m-mlx deployment

36,955 shared input tokens/s

22.72× Jev API · 2.42× same-model PyTorch

Runtime	Latency (ms)	Input tokens/s
322m · MLX	9.10	36,955
322m · PyTorch	22.06	15,247
Jev 1.13 API	206.77	1,627

Local FP16: one A100 40GB · batch/concurrency 1

3 Chinese cases · 60 timed calls/arm · API round trip

12M household application

Largest consumption gains: decision makers aged 25–34.

A higher purchase goal and tied limit add CNY 2,887 consumption per household; closing cash falls CNY 929.

24-month extension

Low-income households most often exhaust cash by month 3.

10,000 households · 2 goals · 5 shocks + reference: persistent income loss drains reserves; sustained gains support larger purchases.

arxiv.org/abs/2610.27746 github.com/KataDavidXD/s2-mas Models + training/evaluation scripts + synthetic companion · Genese implementation private

Genese: execution architecture and measured performance

Brain handles decisions. World/ECS executes effects. Partitions overlap preparation, inference and ledger updates.

Native execution capabilities

Paper Table S13 · 10 mechanisms × 6 frameworks

Capability	Genese	OneSim	OASIS	MiroFish	AgentSociety	MatrAIx
Array-based ECS storage	**Yes**	N/D	N/D	N/D	N/D	N/D
Synchronized state persistence	**Yes**	Yes	No	No	No	No
Native action-rule enforcement	**Yes**	Partial	Yes	Yes	Yes	Partial
Policy-intent / world-effect separation	**Yes**	No	No	No	No	No
Atomic state–event commit	**Yes**	No	No	No	No	No
World-step isolation / rollback	**Yes**	No	No	No	Partial	Partial
Recovery / resume	**Yes**	No	No	No	Yes	Yes
Execution / connection pooling	**Yes**	Yes	No	No	Yes	No
Task admission / flow control	**Yes**	Partial	Yes	Yes	Yes	Yes
Decision / trial concurrency control	**Yes**	Partial	Yes	Yes	Yes	Yes

Yes: implemented Partial: part of the defined mechanism No: absent in the inspected path N/D: not documented

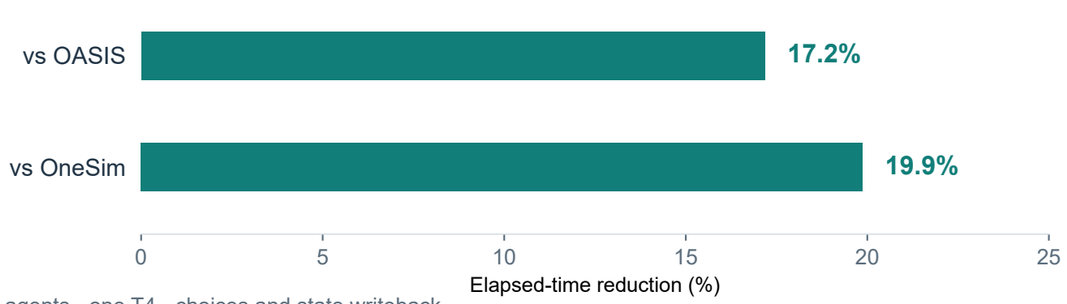
Action rules: OASIS/MiroFish platform rules; AgentSociety permissions and workspace. Resume: AgentSociety checkpoints; MatrAIx job progress.

OneSim: PostgreSQL supports synchronized persistence; event/interface and decision-flow controls are Partial. MatrAIx: questionnaire-output checks are Partial.

AgentSociety and MatrAIx isolate independent tasks; whole-world rollback is a separate mechanism.

17–20% less execution time

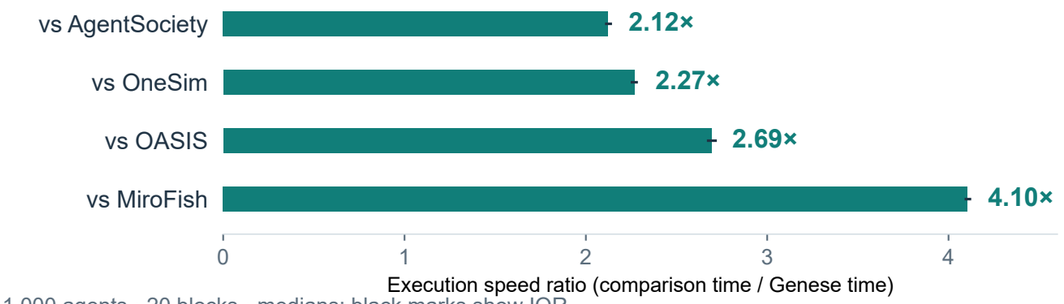
Actual sharp-421m decisions · shared model allocation



1,000 agents · one T4 · choices and state writeback

2.12–4.10× execution speed

Fixed immediate decision backend · relative to each engine



1,000 agents · 20 blocks · medians; black marks show IQR

MatrAIx parallel questionnaire adapter: 1,062.153 s; Genese parallel partitions: 2.743 s on the immediate-backend workload.